

Doug Rynar

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EDUCATION

MEng. Computer Engineering – Lehigh University

2023 – Dec 2026

GPA: 3.92 — Eta Kappa Nu IEEE Honors Society

Coursework: Software Engineering, Computer Architecture, FPGA HLS, Algorithms, Microcontrollers, Digital System Design, Operating Systems, Systems Software, Cryptography

BS. Physics – Lehigh University

2015 – 2019

Coursework: Statistics, Linear Algebra, Electricity & Magnetism, Computational Physics

SKILLS

Programming Languages: C/C++, Python, Verilog, Java

Frameworks and Libraries: Next.js, Ansible, Terraform, Xilinx HLS

Software Tools: Docker, GitLab CI/CD Pipelines, Jira

PROJECTS

Home Monitoring System

- Developed firmware for ESP8266 microcontroller using C for real-time environmental data collection
- Designed and fabricated a PCB sensor board with an I²C and UART interface using KiCad
- Built a containerized Node.js backend SQL database & a React dashboard using Docker for deployment

Social Media App

- Designed and implemented a full stack social media platform in a simulated Agile team environment
- Built a Java API backend with SQL database, developed both Flutter mobile & React web frontends

Discord Music Bot Web App

- Integrated a Discord music streaming bot with a React web UI using Java (Spark framework)
- Leveraged Google Firebase and Firestore for real-time synchronization between frontend & backend
- Designed an interactive music queue and player interface for enhanced user experience

Laboratory Monitoring System Data Validation

- Automated processing of live data from lab stability chambers and room conditions
- Implemented algorithms to detect data gaps, anomalies, and deviations from specifications
- Generated automated reports to notify lab management of potential equipment or data integrity issues

WORK EXPERIENCE

Software Engineer Intern - DevSecOps at Firefly Aerospace SciTec

May 2026 – Present

- Dockerfile CVE scanning pipeline
- AWS machine image compliance with OpenSCAP security tests
- Secret level Security Clearance

Equipment Specialist III - Pace Life Sciences

Mar 2022 – May 2026

- Troubleshooting and repairing analytical lab instrumentation in pharmaceutical GMP environment
- Own the site's schedule for annual Calibration, Preventive Maintenance, and Qualifications
- Ensure compliance of instrumentation/processes with cGMP, ISO 17025, 21 CFR Part 11

Molecular Technologist - LabCorp

Jul 2020 – Feb 2022

- Performed PCR assay with emphasis on instrument maintenance in a collaborative environment